# THE RESOLUTION PROPERTY DESCENDS THROUGH A LIMIT WITH AFFINE TRANSITION MORPHISMS

THE STACKS PROJECT AUTHORS

*Editorial note.* Counted span is the author's own proof environment, source lines 13720-13768, immediately following the statement at 13711-13718.

*Editorial note.* Four cross-chapter pointers (Limits of Spaces, Derived Categories of Spaces, Algebraic Spaces) are recorded in assembly.external_references.

*Editorial note.* The comparison of the two quotient modules is displayed inside the span; the descent lemma is then applied to it.

*Allowed standard prerequisites (editorial).*

- The one-sheaf criterion for the resolution property. chapter lemma lemma-resolution-property-one-sheaf, cited at proof line 13767. Turns the constructed quotient into the resolution property for the approximating space X_i.
- Resolution property goes up along affine morphisms. Derived Categories of Spaces, lemma spaces-perfect-lemma-resolution-property-goes-up-affine, cited at proof lines 13722-13723. Gives the easy direction: X inherits the property from X_i.
- Descending finitely presented modules along a limit. Limits of Spaces, lemma spaces-limits-lemma-descend-modules-finite-presentation, cited at proof lines 13761-13762. Shows the two quotient modules already agree at some stage, which is what produces the quotient on X_i.
- Quasi-compact affine covers and finite type ideals. Limits of Spaces, lemma spaces-limits-lemma-quasi-coherent-finite-type-ideals and Algebraic Spaces, lemma spaces-properties-lemma-quasi-compact-affine-cover, cited in the proof. Produce the affine etale presentation and the finite type ideal sheaf used to build the comparison modules.

*Source reference.* Original lemma 0874: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7806–7806. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 0GV1: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13566–13566. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 07V7: focused-seeds/geometry/S9-spaces-resolve-prop-descends/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

*Date*: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 0855: focused-seeds/geometry/S9-spaces-resolve-prop-descends/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 0GUU: focused-seeds/geometry/S9-spaces-resolve-prop-descends/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 03H6: focused-seeds/geometry/S9-spaces-resolve-prop-descends/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X = \lim X_i$ be a limit of a direct system of quasi-compact and quasi-separated algebraic spaces over $S$ with affine transition morphisms. Then $X$ has the resolution property if and only if $X_i$ has the resolution properties for some $i$.*

*Proof.* If $X_i$ has the resolution property, then $X$ does by Derived Categories of Spaces, Lemma 0GUU. Assume $X$ has the resolution property. Choose $i \in I$. We may choose an affine scheme $V_i$ and a surjective étale morphism $V_i \to X_i$ (Properties of Spaces, Lemma 03H6). We may choose an embedding $j : V_i \to Y_i$ with $Y_i$ finite and finitely presented over $X_i$ (Lemma 0874). We may choose a finite type quasi-coherent ideal $\mathcal{I}_i \subset \mathcal{O}_{Y_i}$ such that $V_i = Y_i \setminus V(\mathcal{I}_i)$ (Limits of Spaces, Lemma 0855). Denote $V \to Y \to X$ the base changes of $V_i \to Y_i \to X_i$ to $X$. Denote $\mathcal{I} \subset \mathcal{O}_Y$ the pullback of the ideal $\mathcal{I}_i$. By the easy direction of Lemma 0GV1 there exists a finite locally free $\mathcal{O}_X$-module $\mathcal{E}$ and a surjection $\mathcal{E} \to \pi_*\mathcal{I}$. Note that since $\pi_i : Y_i \to X_i$ is finite and of finite presentation we also have that $\pi : Y \to X$ is finite and of finite presentation and that the $\mathcal{O}_{X_i}$-modules $\pi_{i,*}\mathcal{O}_{Y_i}$ and $\pi_{i,*}(\mathcal{O}_{Y_i}/\mathcal{I}_i)$ are of finite presentation and pullback to $X$ to give $\pi_*\mathcal{O}_Y$ and $\pi_*(\mathcal{O}_Y/\mathcal{I})$. Thus by Limits of Spaces, Lemma 07V7 after increasing $i$ we can find a finite locally free $\mathcal{O}_{X_i}$-module $\mathcal{E}_i$ and a map $\mathcal{E}_i \to \pi_{i,*}\mathcal{O}_{Y_i}$ whose base change to $X$ recovers the composition $\mathcal{E} \to \pi_*\mathcal{I} \to \pi_*\mathcal{O}_Y$. The pullbacks of the finitely presented $\mathcal{O}_{X_i}$-modules $\mathrm{Coker}(\mathcal{E}_i \to \pi_{i,*}\mathcal{O}_{Y_i})$ and $\pi_{i,*}(\mathcal{O}_{Y_i}/\mathcal{I}_i)$ to $X$ agree as quotients of $\pi_*\mathcal{O}_Y$. Hence by Limits of Spaces, Lemma 07V7 we may assume that these agree, in other words that the image of $\mathcal{E}_i \to \pi_{i,*}\mathcal{O}_{X_i}$ is equal to $\pi_{i,*}\mathcal{I}_i$. Then we conclude that $X_i$ has the resolution property by Lemma 0GV1. □